

# The Proton Color Singlet from Symmetric Register Windows: how tetrahedral $A_4$ symmetry intertwines the color $\mathbb{Z}_3$ at the $W_{ABC}$ junction

Tessera — cobordism subsystem (#398, completing #396)

## Abstract

The trivalent junction  $W_{ABC}$  (#396) carries the  $\mathbb{Z}_3$  shadow of the three-quark color singlet  $\varepsilon_{ijk}$  that the bipartite merge cannot (#382): three input “windows”  $A, B, C$  and one emergent result window  $R$ , each a triple of holes, all on one connected 2-frequency geodesic icosahedron ( $S^2$ , 42 vertices, 80 triangles) extruded  $\times I$ . After #396 the singlet was *reachable* — the input  $\rightarrow$  result transport is full-rank with the singlet in its image — but a *natural* quark input reached only  $\sim 0.74$ : the greedily placed windows are geometrically *inequivalent*, so the per-window transport blocks are unrelated matrices. This note records the principled fix and a finding that was not anticipated. We place the four windows as *one orbit of a tetrahedral subgroup*  $A_4$  of the icosahedral rotation group, seated at a system of imprimitivity on the icosahedron’s vertices. We then show that symmetric windows are *necessary but not sufficient*: the singlet emerges only when three conditions hold together — (i) a symmetry-respecting (uniform) metric, (ii) the input in the  $\omega$ -representation of the window-cycling symmetry  $g$ , and (iii) a  $g$ -coherent signed read-out. Given these, the transport  $M$  *intertwines* the color  $\mathbb{Z}_3$ ,  $MP_{\text{in}} = P_{\text{out}}M$ , and the  $\omega$ -representation input transports to the singlet line with overlap 0.999 — the proton, forced by symmetry and never imposed. The residual  $\sim 4 \times 10^{-2}$  in the intertwining is a fixed-resolution triangulation artifact (the  $\times I$  extrusion is not  $g$ -symmetric even though the base surface and metric are); it is metric-independent, and 0.999 rather than an exact 1 is its signature. All numbers are measured through the production C++ `TransportCobordism`.

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## 1 Where we were

The proton-from-quarks program builds the baryon color singlet by relaxing a Lorentzian Regge cobordism with pinned boundary color states and reading the *emergent* result off the relaxed geometry — nothing is imposed. We write the *color charge* of a window  $W$  for the net signed period it carries,

$$\sigma_W = \sum_{\text{holes } h \in W} (\text{signed induced period of } h), \quad (1)$$

the discrete analogue of  $\sum_i \psi_i$  on a color triple. Two prior results frame the present one.

**The bipartite obstruction (#382).** Composing pairwise merges never reaches the singlet: the emergent result stays color-charged ( $|\sigma| \approx 0.67$  for the tripartite sequence) at every step. This is representation theory, not a numerical defect: the color singlet  $\varepsilon_{ijk}$  is the three-index alternating invariant, absent from  $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$  and first appearing in  $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$ . A baryon from fundamentals is irreducibly an  $n = 3$  object.

**The tripartite junction (#396).** The simplicial resolution is the trivalent junction  $W_{ABC}$ : one connected base surface — a 2-frequency geodesic icosahedron ( $S^2$ , 42 vertices, 80 triangles) — minus 12 vertex-disjoint hole triangles grouped into four *windows* of three,  $A, B, C$  (the quark inputs) and  $R$  (the emergent result), extruded  $\times I$  into a 3-complex (**prismCells**). The 12 hole cycles satisfy exactly one relation — the global Stokes constraint below — so  $b_1 = 12 - 1 = 11$ ; the inputs are *independent* cycles (they do not average, escaping the bipartite obstruction), and that single relation is junction charge conservation:

$$\sum_{\text{all holes}} (\text{induced period}) = 0 \quad \implies \quad \sigma_R = -(\sigma_A + \sigma_B + \sigma_C). \quad (2)$$

Confinement is then a property of (2): the four window charges must sum to zero, so an isolated colored input cannot close on the junction; only the color-neutral (singlet) combination does. The bipartite  $|\sigma| \approx 0.67$  residual is precisely such a non-confined, charged failure.

**The open problem.** After #396 the read-out is a linear map  $M : \mathbb{C}^9 \rightarrow \mathbb{C}^3$  (nine input color amplitudes, three result amplitudes); it is full-rank ( $\text{rank } M = 3$ ) and the singlet lies in its image, so the singlet is *reachable*. Yet the natural neutral input (three identical color-neutral pairs, read with each hole's own orientation) gives overlap  $\sim 0.74$ . The diagnosis: with the windows placed *greedily* (first twelve vertex-disjoint triangles in face order) the per-window transport blocks  $M_A, M_B, M_C$  (the three  $3 \times 3$  sub-blocks of  $M$ ) are geometrically inequivalent —  $M_A^{-1} M_B$  is a general matrix, not a signed permutation. The input that hits the singlet is therefore geometry-specific and unnatural. The fix is to make the windows *equivalent*, which is the subject of this note.

## 2 Symmetric placement: the windows as an $A_4$ orbit

### 2.1 The symmetry available

The regular icosahedron has rotation group  $I \cong A_5$  of order 60; the 2-frequency geodesic subdivision inherits it as a group of simplicial automorphisms permuting the 80 sub-triangles and fixing the metric of the round sphere.  $A_5$  contains tetrahedral subgroups  $T \cong A_4$  of order 12. Fix one such  $A_4$ . Because  $\gcd(|A_4|, 5) = 1$  and the vertex stabilizer in  $A_5$  is the 5-fold rotation  $C_5$ ,  $A_4$  acts *freely and transitively* on the 12 vertices (one orbit of 12). It nevertheless admits a *system of imprimitivity* with four blocks of three,

$$\{2, 8, 10\}, \quad \{1, 4, 7\}, \quad \{0, 6, 9\}, \quad \{3, 5, 11\}, \quad (3)$$

which partition the twelve vertices;  $A_4$  permutes the four blocks by its standard transitive degree-4 action, and each block is fixed setwise by one of  $A_4$ 's four  $C_3$  subgroups (its three-fold axis).

### 2.2 The construction

A *window* is the triple of vertex-disjoint corner sub-triangles seated at one block of (3) and cycled by that block's three-fold axis. Concretely the corner sub-triangle at icosahedron vertex  $v$  is  $(v, m(v, p), m(v, q))$ , where  $m(\cdot, \cdot)$  is the edge-midpoint vertex and  $p, q$  are two neighbours of  $v$ ; the  $C_3$  rotation about the block's axis cycles the three corners, hence the three holes. The four windows  $W_A, W_B, W_C, W_R$  are therefore a *single*  $A_4$  orbit (their stabilizer is the block's own  $C_3$ , and  $|A_4|/3 = 4$ ). All 12 holes are vertex-disjoint (the 36 vertices are distinct), and the topology is unchanged:  $b_1 = 11$  and the dual complex passes the manifold gate.

**Proposition 1** (Symmetric windows). *The four windows in the cobordism's own vertex labels are*

$$\begin{aligned} W_A : & (2, 13, 14) (8, 31, 32) (10, 22, 36) & W_B : & (1, 23, 24) (4, 19, 34) (7, 30, 39) \\ W_C : & (0, 16, 18) (6, 26, 27) (9, 33, 41) & W_R : & (3, 15, 29) (5, 20, 21) (11, 37, 38) \end{aligned}$$

*each window a genuine  $C_3$  orbit, all real faces of the base surface, the 36 vertices distinct.*

**Generated, not hardcoded.** The windows are produced *from the symmetry* at build time: four  $C_3$  generators (as 12-vertex permutations), one seed corner per window, and the midpoint lift (a base vertex maps by the permutation; a midpoint  $m(p, q)$  maps to  $m(p', q')$ ). This is correct in whatever midpoint numbering the build produces. The point is not pedantic: two independent attempts to *hardcode* the twelve triples both produced tables that were wrong against the actual numbering. Generation is correct-by-construction and is guarded by build-time assertions (each window a  $C_3$  orbit, all holes real faces, 36 distinct vertices). The explicit data are in Appendix A.

## 3 Necessary but not sufficient: three ingredients

The central empirical finding is that symmetric *windows alone do not produce the singlet*. Three conditions must hold together — a symmetry-respecting metric, the input in the right representation, and a coherent read-out — and all three are phrased in terms of the same window-cycling symmetry, which we name first.

**Definition 1** (The window-cycling symmetry  $g$ ).  $A_4$  acting on the four windows is the alternating group on four points; its three-cycles fix one window and cycle the other three. Let  $g \in A_4$  be the element with

$$g : W_R \mapsto W_R, \quad W_A \mapsto W_B \mapsto W_C \mapsto W_A. \quad (4)$$

Such  $g$  exists and is unique up to its inverse; on the result window it acts as a  $C_3$  on  $R$ 's three holes. *This  $g$  is the color  $\mathbb{Z}_3$ .*

We write  $P_{\text{in}}$  (resp.  $P_{\text{out}}$ ) for the signed permutation by which  $g$  acts on the nine input-hole (resp. three result-hole) period amplitudes; both are made precise in §4.

### 3.1 (i) A symmetry-respecting metric

The seed metric must be invariant under  $g$ . A random “jitter” seed — previously used to lift the degenerate uniform point — breaks the  $A_4$  symmetry: charge conservation (2) degrades from  $\sim 10^{-14}$  to  $\sim 0.07$ , and no symmetric input reaches the singlet. The default seed is now the *uniform* metric  $l^2 = 1$ , which is manifestly  $g$ -invariant; on it (2) is exact ( $|\sigma_R| \sim 10^{-14}$  for a symmetric neutral input). The mutual-information (van Raamsdonk) seed, `setEntangledMetric` — entanglement-derived edge lengths that are constant on  $g$ -orbits, hence  $g$ -invariant — works equally well.

Crucially, the singlet is read from the *transport* at the symmetric seed, not from a relaxation: the intertwining of §4 holds already at iteration zero. Relaxation still runs and is well-behaved (60 iterations reduce the stationarity residual from 204 to 8.4), but it is not what produces the singlet.

### 3.2 (ii) The $\omega$ -representation input

The “natural symmetric quark input” is *not*  $[1, \omega, \omega^2]$  placed identically on  $A, B, C$  (that gives only  $\sim 0.48$ ). It is the input that transforms in the  $\omega$ -representation of  $g$ :

$$P_{\text{in}} \phi = \omega \phi, \quad \omega = e^{2\pi i/3}. \quad (5)$$

Physically: the three quark colors carry the correct relative  $\mathbb{Z}_3$  phase under the geometric cycling, not the same vector repeated. The input representation is fixed by  $g$  — a property of the construction — *before* any output is computed; the theorem’s content (§4) is that the transport carries it to the singlet line rather than scrambling it, as the failed rows of §5 show it does for non- $\omega$ -rep inputs.

### 3.3 (iii) A $g$ -coherent signed read-out

The period of a 1-cochain around a hole triangle is orientation-dependent. The read-out must use orientations coherent under  $g$ : the period of each hole is taken in the orientation propagated from a chosen base hole by  $g$ , i.e. multiplied by the orientation parity  $\varepsilon$  of  $g$  on it (the corresponding entry of  $P$ ). Equivalently, on the three holes of  $R$  the read-out signs are the entries of  $P_{\text{out}}$ . The default per-window covector is *not*  $g$ -coherent — its signs are inconsistent across windows ( $[-1, -1, 1], [1, -1, -1], [1, -1, 1]$ ) — and with raw sorted periods the symmetry is broken in the read-out and the singlet is missed.

## 4 The intertwining theorem

Let  $M : \mathbb{C}^9 \rightarrow \mathbb{C}^3$  be the input→result transport:  $M$  sends the nine pinned input-hole periods to the three result-hole periods of the carried harmonic.  $M$  is linear because the carried-representative map is linear.

**Lemma 1** (Equivariance on the base surface). *On the base surface (the holed geodesic sphere),  $g$  is a metric-preserving simplicial automorphism. It acts on hole-period vectors by a signed permutation  $P$ , the sign on a hole being the orientation parity of  $g$  on it;  $P = P_{\text{in}} \oplus P_{\text{out}}$  because  $g$  maps input holes to input holes and result holes to result holes. The base-surface carried map commutes with  $g$ :*

$$M P_{\text{in}} = P_{\text{out}} M. \quad (6)$$

*Proof.*  $g$  preserves edge lengths, so it preserves the discrete Hodge inner products and commutes with the coboundary and Hodge Laplacian; the least-norm harmonic realizing a set of pinned periods is carried to the harmonic realizing the  $g$ -transported periods. A hole  $(a, b, c)$  (oriented, sorted) has period  $\psi_{ab} + \psi_{bc} - \psi_{ac}$ , invariant under cyclic relabelling and negated under a transposition;  $g$  sends  $(a, b, c) \mapsto (ga, gb, gc)$ , whose sorted form differs by a permutation of parity  $\varepsilon \in \{\pm 1\}$ , so the period maps with sign  $\varepsilon$ . Collecting the signs gives  $P = P_{\text{in}} \oplus P_{\text{out}}$  and (6).  $\square$

**Theorem 1** (The  $\omega$ -input transports to the singlet). *Assume the exact equivariance (6). Then  $P_{\text{out}}$  is a signed 3-cycle with eigenvalues  $\{1, \omega, \omega^2\}$ , and its  $\omega$ -eigenspace is the one-dimensional color singlet line  $\mathbb{C}[1, \omega, \omega^2]$  in  $R$ 's hole basis. For any input  $\phi$  with  $P_{\text{in}}\phi = \omega\phi$ , the result  $M\phi$  lies in that singlet line. Moreover  $M$  is rank 3 with  $\phi \notin \ker M$  (so  $M\phi \neq 0$ ), and the  $\omega$ -eigenspace of  $P_{\text{in}}$  is three-dimensional, so a three-parameter family of symmetric inputs all transport to the singlet.*

*Proof.*  $g$  cycles  $R$ 's three holes, so  $P_{\text{out}}$  is a signed 3-cycle; the product of its three signs is  $+1$  (verified: the eigenvalue angles are  $\{0, \pm 2\pi/3\}$ ), hence its eigenvalues are the cube roots of unity, each eigenspace one-dimensional. The  $\omega$ -eigenvector is the discrete Fourier mode  $[1, \omega, \omega^2]$  — the color singlet. For  $P_{\text{in}}\phi = \omega\phi$ , equation (6) gives  $P_{\text{out}}(M\phi) = M(P_{\text{in}}\phi) = \omega(M\phi)$ , so  $M\phi$  is an  $\omega$ -eigenvector of  $P_{\text{out}}$ , i.e.  $M\phi \in \mathbb{C}[1, \omega, \omega^2]$ ; and  $M\phi \neq 0$  since  $M$  is rank 3 and  $\phi$  is not in its kernel. Finally  $P_{\text{in}}$  is three disjoint signed 3-cycles (each an  $A$ -hole, its  $B$ -image, its  $C$ -image), each with sign-product  $+1$ , so  $\omega$  occurs with multiplicity three.  $\square$

**Remark 1** (The measured object is equivariant only up to a triangulation residual).  $g$  is a metric-preserving automorphism of the base surface, but the cobordism is the base  $\times I$ , and the prism tetrahedralization's diagonal choices are not  $g$ -invariant —  $g$  permutes vertices but not simplices. Hence the carried map *through* the 3-complex is only approximately equivariant,  $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\| \approx 4.3 \times 10^{-2}$  (§5), and the  $\omega$ -input transports to the singlet line up to that residual — overlap 0.999 rather than exactly 1. The residual is metric-independent (§5); a  $g$ -symmetric subdivision of the extrusion would remove it.

**Remark 2** (Manifest  $\mathbb{Z}_3$ , and the relation to  $S_3$ ). What is geometrically realized is the color  $\mathbb{Z}_3$ :  $g$  is its generator, the transport intertwines it, and the singlet is its  $\omega$ -Fourier mode — the  $\mathbb{Z}_3$  mode that survives restricting the totally antisymmetric  $\varepsilon_{ijk}$  to the cyclic subgroup. The full color Weyl group is  $S_3$ ; its  $\mathbb{Z}_3$  cyclic part is the one made manifest by the tetrahedral geometry, and the conjugate symmetric input ( $P_{\text{in}}\phi = \omega^2\phi$ ) lands on the conjugate singlet  $[1, \omega^2, \omega]$ . The singlet is thus not a tuned target but the forced image of an entire eigenspace — the sense in which it is emergent rather than imposed.

## 5 Numerical results

All figures are measured through the production C++ `TransportCobordism` with the symmetric `TripartiteRegisterTopology`; the symmetry  $g$  and the signed reps  $P_{\text{in}}, P_{\text{out}}$  are reconstructed from the icosahedral generators applied to the cobordism’s *own* window labels (never Python-reconstructed holes). The figure of merit is the normalized overlap

$$\text{overlap} = \frac{|\langle v_R, s \rangle|}{\|v_R\| \|s\|}, \quad s = [1, \omega, \omega^2], \quad (7)$$

where  $v_R$  is the carried result-hole period vector (read from  $R$  after the carry, never pinned).

| Placement / metric / input                                                 | overlap with $s$ | note                            |
|----------------------------------------------------------------------------|------------------|---------------------------------|
| greedy windows, jitter, natural neutral input                              | $\sim 0.74$      | windows inequivalent            |
| symmetric windows, uniform, $[1, \omega, \omega^2]$ identical on $A, B, C$ | 0.48             | wrong input (not $\omega$ -rep) |
| symmetric windows, uniform, naive neutral, raw read                        | 0.65             | wrong (non-coherent) read-out   |
| <b>symmetric windows, uniform, <math>\omega</math>-rep input</b>           | <b>0.999</b>     | the proton                      |

The structural and symmetry measurements:

- topology:  $b_1 = 11$ , dual complex valid (a manifold, no weld); metric uniform.
- $P_{\text{out}}$  eigenvalues  $\{1, \omega, \omega^2\}$  (angles  $\{0, \pm 2\pi/3\}$ ): the color  $\mathbb{Z}_3$ .
- intertwining  $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\| = 4.26 \times 10^{-2}$ .
- the three  $\omega$ -eigenvectors of  $P_{\text{in}}$  map to the singlet with overlaps 0.9991, 0.9998, 0.9991; the small spread is the fixed-resolution triangulation artifact (basis-dependent), with 0.999 the reported floor.
- charge conservation (2) exact ( $10^{-14}$ ) for a symmetric neutral input on the uniform metric; the singlet is traceless,  $1 + \omega + \omega^2 = 0$ .

**The residual is a triangulation artifact, not symmetry breaking.** The intertwining is not exactly zero, and the cause is *not* the metric: replacing the uniform seed by a  $g$ -invariant *non-degenerate* metric (edge lengths constant on  $g$ -orbits) gives the same residual,  $4.19 \times 10^{-2}$  versus  $4.26 \times 10^{-2}$ . The base surface, its holes, and the metric are all  $g$ -symmetric; what is not is the prism tetrahedralization (the  $\times I$  extrusion’s diagonal choices), so the discrete carried map is only approximately equivariant at this fixed resolution (Remark 1). The 0.999 overlap, not the exact 1, is the signature of that artifact; a  $g$ -symmetric subdivision would remove it. This is the conclusive figure that the symmetry is respected.

## 6 Implementation

The change is confined to `TripartiteRegisterTopology`.

- `geodesicTwoSphere` now returns its edge-midpoint map alongside the faces, so the window generator can lift the  $C_3$  permutations.
- `symmetricWindows` generates the four windows from the four  $C_3$  generators, a seed corner per window, and the midpoint lift, with the build-time assertions of Proposition 1; it replaces the greedy twelve-hole pick.

- the default metric seed is uniform  $l^2 = 1$  (the jitter seed is removed); the van Raamsdonk seed (`setEntangledMetric`) remains as a  $g$ -invariant alternative.

The result block  $R$  still *emerges* — it is never pinned or hand-placed: the read-out pins the three inputs on  $A, B, C$  and returns  $R$  to be read off the transport after the carry. Validation reconstructs  $g$  and  $P_{\text{in}}, P_{\text{out}}$  from the cobordism’s own labels and checks Theorem 1 end-to-end. A worked demonstration is `examples/cobordism/proton_tripartite_junction.py`.

## 7 Discussion

The proton singlet is now *forced by symmetry*. With symmetric windows and a symmetric metric the transport intertwines the color  $\mathbb{Z}_3$ ; the color-symmetric quark input is, by definition, the  $\omega$ -representation; and any such input lands on the unique  $\omega$ -eigenvector of the result, the singlet  $[1, \omega, \omega^2]$ . Nothing about the singlet is inserted: it is the image of an entire eigenspace, and a three-parameter family of symmetric inputs all reach it. The earlier 0.74 ceiling was not a limitation of the geometry but of *inequivalent* windows; the earlier 0.48 and 0.65 “failures” were the wrong input and the wrong read-out, not the wrong construction.

Two honest caveats. First, the geometrically realized symmetry is  $\mathbb{Z}_3$ , the cyclic subgroup of the color Weyl group  $S_3$ ; the cyclic singlet is what is forced. Second, the intertwining is exact only up to a fixed-resolution triangulation artifact ( $\sim 4 \times 10^{-2}$ ), so the measured overlap is 0.999 rather than 1 (Remark 1).

The path forward (#380) reads emergent observables off the relaxed singlet geometry — net charge, a radius, the mass ratio — and follows the photon (a worldline whose  $l^2$  relaxes through null) into re-absorption.

## 8 Conclusion

The trivalent junction with *symmetric* register windows resolves the last gap in building the proton from quarks. Placing the four windows as one tetrahedral  $A_4$  orbit makes them equivalent; on a symmetry-respecting metric the input→result transport intertwines the color  $\mathbb{Z}_3$ ; and the natural color-symmetric input then transports to the color singlet with overlap 0.999. Symmetric windows are necessary but not sufficient — the metric, the input representation, and the read-out must all respect the same symmetry — and when they do, the proton’s color singlet is not assembled but *compelled*.

## A Explicit data

The four  $C_3$  generators as 12-vertex permutations (image of vertex  $i$  at position  $i$ ), each cycling its window’s vertex block:

$$\begin{aligned} W_A (2 \rightarrow 8 \rightarrow 10) : & \quad [4, 3, 8, 9, 5, 0, 7, 11, 10, 1, 2, 6] \\ W_B (1 \rightarrow 4 \rightarrow 7) : & \quad [3, 4, 0, 2, 7, 8, 5, 1, 6, 11, 9, 10] \\ W_C (0 \rightarrow 6 \rightarrow 9) : & \quad [6, 10, 11, 7, 2, 1, 9, 8, 3, 0, 5, 4] \\ W_R (3 \rightarrow 5 \rightarrow 11) : & \quad [10, 6, 1, 5, 9, 11, 2, 0, 4, 8, 7, 3] \end{aligned}$$

The seed corner of window  $w$  is  $(v, m(v, n_1), m(v, n_2))$  with  $(v; n_1, n_2)$  equal to

$$(2; 0, 1), \quad (1; 6, 10), \quad (0; 3, 4), \quad (3; 2, 7)$$

respectively; the window is the  $C_3$  orbit of its seed. The window-cycling element  $g$  of Definition 1 is the  $A_4$  rotation whose induced window permutation is  $W_A \rightarrow W_B \rightarrow W_C \rightarrow W_A$  with  $W_R$  fixed; it acts on  $R$ 's three holes as a 3-cycle, giving  $P_{\text{out}}$  its eigenvalues  $\{1, \omega, \omega^2\}$ .